

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO3: *Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 30%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

CODING CHALLENGE

Duration: 2hrs

Code of Conduct:

I ALLAPAREDDY JAHNAVI (192472214) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

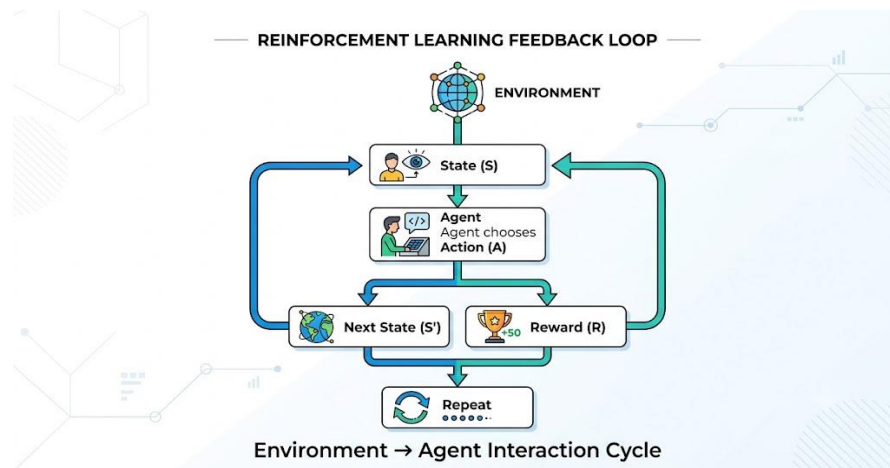
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A. Jahnavi

1) Predict the Reinforcement Learning (RL) framework by describing the agent–environment interaction, emphasizing the importance of states, actions, and rewards, and discuss how cumulative reward influences decision-making.

Reinforcement Learning (RL) is a learning paradigm where an **agent** interacts with an **environment** to achieve a goal by maximizing cumulative rewards.

RL Framework



Components

Agent

- Learns from experience.
- Chooses actions.

Environment

- Everything outside the agent.
- Responds to agent actions.

State (S)

- Represents the current situation.

Example:

Robot position = (2,3)

Action (A)

Possible actions include:

- Up
- Down
- Left

- Right

Reward (R)

Feedback received after every action.

Example:

Reach Goal $\rightarrow +10$

Hit Obstacle $\rightarrow -5$

Normal Move $\rightarrow -1$

Cumulative Reward

Instead of maximizing immediate reward, RL maximizes

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

where

- γ = Discount Factor

A higher cumulative reward results in better long-term decisions rather than greedy short-term choices.

Conclusion

The RL framework enables agents to continuously improve decision-making through interaction with the environment while maximizing long-term cumulative rewards.

2) Estimate how decision-making problems can be modeled as a Markov Decision Process (MDP), detailing the components of action space, transition probability, reward function, and discount factor.

A Markov Decision Process (MDP) provides a mathematical framework for solving sequential decision-making problems.

An MDP consists of the tuple:

$$(S, A, P, R, \gamma)$$

Components

1. State Space (S)

Set of all possible situations.

Example:

Robot positions, Traffic conditions, Game states

2. Action Space (A)

Possible actions available.

Example:

Move Left

Move Right

Move Up

Move Down

3. Transition Probability (P)

Probability of moving to another state.

$$P(s' | sa)$$

Example

Move Right

90% → Desired state

10% → Slip downward

4. Reward Function (R)

Immediate feedback after taking an action.

Example

Goal reached = +100

Obstacle = -20

Step cost = -1

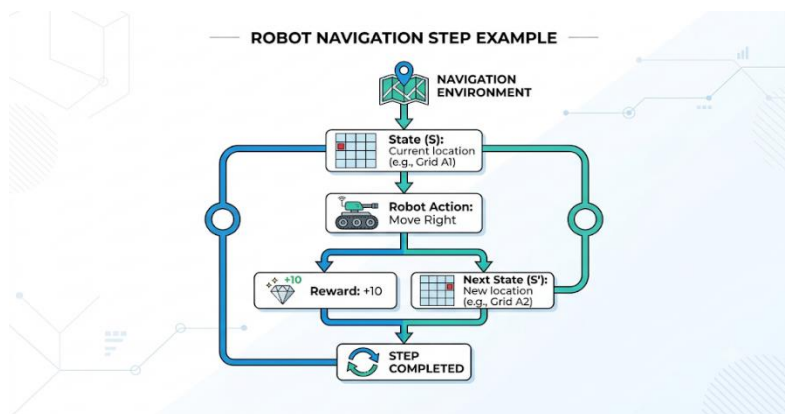
5. Discount Factor (γ)

$$0 \leq \gamma \leq 1$$

$\gamma = 0$, Immediate rewards only.

$\gamma = 1$, Future rewards equally important.

Example



Conclusion

MDPs allow RL problems to be represented mathematically and help agents make optimal sequential decisions.

3) Discuss the role of policy and value functions in RL, and examine how state-value and action-value functions contribute to evaluating long-term benefits of actions using the Bellman Equation.

Policy (π)

A policy determines which action an agent should choose.

$$\pi(a | s)$$

It maps

State $\rightarrow$ Action

Value Function

Measures how good a state is.

$$V(s)$$

Higher value



Better future rewards

Action-Value Function

$$Q(s, a)$$

Measures the usefulness of taking action **a** in state **s**.

Bellman Equation

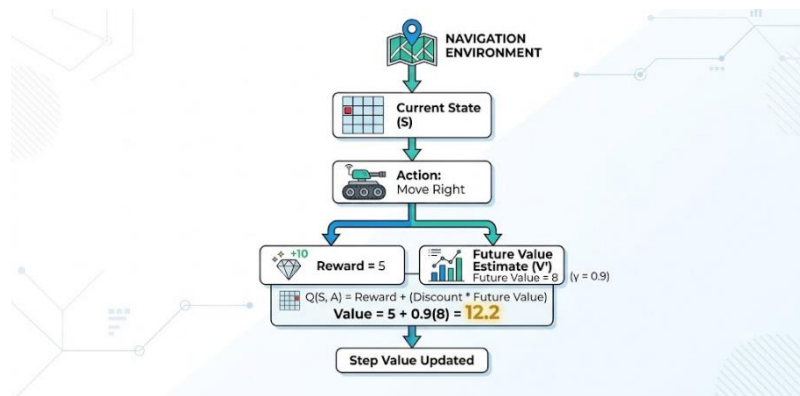
State Value

$$V(s) = R + \gamma V(s')$$

Q-value

$$Q(s, a) = R + \gamma \max_{a'} Q(s', a')$$

Example



Importance

- Evaluates long-term benefits.
- Helps choose optimal actions.
- Forms the basis of dynamic programming and Q-learning.

4) Evaluate the effectiveness of model-free RL compared to model-based RL in dynamic environments, and differentiate their strengths and limitations.

Model-Free RL

Learns directly through interaction.

Examples

- Q-learning
- SARSA

Advantages

- Easy to implement
- Works when environment model is unknown
- Suitable for complex environments

Disadvantages

- Requires many training samples
- Slower convergence

Model-Based RL

Agent first learns the environment model.

Advantages

- Faster learning
- Better planning
- Requires fewer interactions

Disadvantages

- Difficult to model complex environments
- Model inaccuracies reduce performance

Comparison

Feature	Model-Free RL	Model-Based RL
Environment Model	Not Required	Required
Learning Speed	Slower	Faster
Planning	No	Yes
Sample Efficiency	Low	High
Implementation	Simple	Complex

Conclusion

Model-free RL is suitable for unknown environments, whereas model-based RL performs better when an accurate model is available.

5) Illustrate with an example how Q-learning updates the action-value function during training, and demonstrate its convergence behavior.

Q-learning is an off-policy reinforcement learning algorithm that learns the optimal action-value function.

Update Equation

$$Q(s, a) = Q(s, a) + \alpha[R + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

where

α = Learning Rate

γ = Discount Factor

Example

Current Q-value

$$Q = 5$$

Reward

$$R = 10$$

Future Maximum Q

$$\max Q = 8$$

Learning Rate

$$\alpha = 0.5$$

Discount

$$\gamma = 0.9$$

Calculation

Target

$$10 + 0.9(8) = 17.2$$

Updated Q

$$\begin{aligned} Q &= 5 + 0.5(17.2 - 5) \\ Q &= 11.1 \end{aligned}$$

Thus, the Q-value improves after receiving new experience.

Convergence

As the agent repeatedly explores the environment:

- Q-values become more accurate.
- Updates become smaller.
- Q-values converge to the optimal action-value function.
- The learned policy approaches the optimal policy.

Applications

- Robot navigation
- Game playing
- Autonomous driving
- Route optimization

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately